

Aryan Sahni

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SUMMARY

Research-driven Computer Science undergraduate at VIT Chennai with expertise in Artificial Intelligence, Machine Learning, Systems Programming, and Full-Stack Development. Hands on experience fine-tuning Veena TTS using Indigo call-center data and designing a low-latency streaming architecture for a production-ready TTS service. Experienced in building full-stack AI platforms, Graph-RAG systems, and LLM-powered applications using TensorFlow, ONNX Runtime, LangChain, and modern AI frameworks. Published author with research in medical image analysis (ImageCLEF 2025, CEUR Workshop Proceedings) and quantum machine learning (ICESIC 2026), focused on developing scalable, data-driven AI solutions that bridge cutting-edge research with real-world applications.

EDUCATION

Vellore Institute of Technology, Chennai
B.Tech in Computer Science and Engineering

2023–2027
CGPA: 9.07/10

WORK EXPERIENCE

Data Science & AI Research Intern — AIONOS

[\[AIONOS\]](#)

- Designed a low-latency streaming framework for a **3B-parameter LLM-based Text-to-Speech (TTS)** system, reducing **Time-to-First-Audio (TTFA)** by **92%** (19.43 s → 1.55 s) using **KV-cache incremental generation, chunk-wise SNAC decoding**, and streaming audio reconstruction.
- Designed and automated the end-to-end data preparation pipeline for **2000+ minutes** of in-house airline call recordings, including annotation processing, audio pre-processing, **SNAC-tokenization**, and training data generation for **QLoRA-based domain adaptation**.
- Developed a comprehensive benchmarking framework with **232 multilingual evaluation utterances**, automating latency, speech quality (**WER/CER**), speaker similarity, and GPU performance evaluation across baseline, fine-tuned, and streaming TTS models.

PROJECTS

EMAIL Bait Mail

[\[GitHub\]](#)

- Built a **Chrome extension** for real-time **phishing and scam detection** leveraging the **Gmail API, Flask microservices**, and DOM event tracking via **Mutation Observer**.
- Fine-tuned an **AWS Bedrock Foundation Model** using a curated **phishing and scam email dataset** to improve threat classification reliability, enabling **LLM-powered email intent analysis**, phishing detection, and contextual response generation.
- Implemented asynchronous front-end/back-end communication pipelines and dynamic content parsing for low-latency inbox monitoring and real-time email threat analysis.

Graph-RAG Knowledge Retrieval System

[\[GitHub\]](#)

- Built an end-to-end **Graph-RAG pipeline** using **LangChain, Ollama, and Zephyr LLMs**, achieving **78.8% response faithfulness** with **77% effective context utilization**.
- Enhanced retrieval quality through **knowledge graph traversal** and multi-hop reasoning, reducing **hallucinations to 12.6%** and improving factual grounding of generated responses.
- Developed an automated benchmarking framework for **faithfulness, hallucination, graph coverage**, and **LLM-as-a-Judge evaluation**, enabling reproducible performance analysis across Graph-RAG configurations.

SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, Bash

Tools & Platforms: Git/GitHub, Linux, Docker, ONNX Runtime

Frameworks & Libraries: TensorFlow, PyTorch, LangChain, Hugging Face Transformers, Pandas, NumPy, Scikit-learn

AI/ML Expertise: Generative AI, Large Language Models (LLMs), Speech AI & TTS, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), Graph-RAG, Reinforcement Learning, Model Fine-Tuning, Model Evaluation

PUBLICATIONS

Sahni, Aryan, Rachit Gupta, et al. (2025). “Evaluating deep CNNs for multi-label concept detection in ROCov2 radiology image dataset by team LekshmiscopeVIT”. In: *CLEF2025 Working Notes, CEUR Workshop Proceedings, CEUR-WS.org, Madrid, Spain*. URL: https://ceur-ws.org/Vol-4038/paper_203.pdf.

Sahni, Aryan, Krishnesh Mathur, et al. (2026). “An Optimal Evaluation of Entanglement Depth in Variational Quantum Classifiers for Efficient Deployment of Machine Learning Applications”. In: *2026 International Conference on Electronic Systems and Intelligent Computing (ICESIC)*, pp. 320–325. DOI: [10.1109/ICESIC67389.2026.11496417](https://doi.org/10.1109/ICESIC67389.2026.11496417).